PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

Team 166_EyeScream (Domain 1)

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Project Name: TaxMate

Version: 2.0

1. Project Overview

- **Problem Statement:** Malaysia's 140,000+ SMEs face a persistent tax compliance crisis that existing tools have failed to solve.

  For most SME owners in the RM1M–5M revenue bracket, tax compliance is not a structured process — it is a monthly scramble. Receipts pile up in shoeboxes, coordination with accountants happens over WhatsApp and email, and the fear of filing errors looms every reporting cycle. Most owners lack the financial literacy to distinguish claimable from non-claimable expenses, yet are too small to afford an in-house finance team. Compliance ends up entirely dependent on a single external accountant — a fragile, expensive, and unscalable arrangement that leaves SMEs exposed to LHDN penalties at every step.

  On the accountant side, 60–70% of working hours are consumed by low-value data entry and document chasing, leaving little room for the advisory work they are trained to do. This bottleneck slows the entire compliance pipeline for both parties.

  The burden is only growing. Beyond SST, Malaysia's expanding digital tax agenda — including the mandatory E-Invoicing rollout under MyInvois — is introducing new layers of technical compliance that most SMEs are wholly unprepared for. No existing solution addresses this end-to-end: generic accounting software demands heavy manual input, full-service firms are cost-prohibitive, and point solutions leave the owner-accountant coordination gap entirely unresolved.

  TaxMate is built to close that gap — starting with SST compliance today, and expanding across Malaysia's broader digital tax compliance landscape as the regulatory environment evolves.

- **Target Domain:** AI Systems & Agentic Workflow Automation — specifically targeting end-to-end tax compliance workflow automation for Malaysian SMEs, bridging business owners and accounting professionals through an intelligent, GLM-powered platform.

- **Proposed Solution Summary:** TaxMate transforms the fragmented tax compliance process into an intelligent, automated workflow using GLM as the central reasoning engine. Business owners upload receipts and invoices in any format; TaxMate's AI agents automatically extract structured data via OCR, classify document types, detect anomalies, and generate SST-02 draft reports — with the architecture designed to extend towards full MyInvois E-Invoice compliance as the platform scales. Accountants receive a prioritised review queue where they can verify AI-extracted results and digitally sign off — all within a single platform. If the GLM component is removed, TaxMate becomes a mere file storage tool; the AI is what makes the workflow intelligent, adaptive, and scalable.

2. Background & Business Objective

- **Background of the Problem:** For decades, tax compliance in Malaysia has followed a rigid, manual workflow. Business owners collect physical receipts and invoices, manually key in figures into spreadsheets, and send documents to accountants via WhatsApp or email. Accountants then spend hours cross-referencing entries, verifying SST classifications, and preparing submission-ready reports — all by hand. Most SMEs currently have no structured system to manage this process, and as Malaysia's digital tax requirements expand — including the progressive rollout of mandatory E-Invoicing under LHDN's MyInvois portal — the gap between what SMEs can handle manually and what compliance now demands is widening rapidly.

- **Importance of Solving This Issue:** Malaysia has over 140,000 registered SMEs that collectively file SST returns every two months, with each business spending an estimated 8–15 hours per submission cycle on manual document handling alone. At the same time, a traditional accountant can realistically serve only 8–12 SME clients per month, bottlenecked by repetitive data entry and document verification. By automating the end-to-end compliance workflow with GLM-powered agents, TaxMate compresses what once took days into minutes — reducing human error, lowering compliance costs, and enabling accountants to shift from data processors to decision approvers, potentially serving 3–4

times more clients. As Malaysia's digital tax obligations grow beyond SST, the cost of not solving this problem compounds with every filing cycle.

Key impact metrics: 140,000+ SMEs filing SST every 2 months; 8–15 hours of manual work per cycle reduced to under 30 minutes; accountant capacity growing from 8–12 clients to 30–40 clients per month;

- **Strategic Fit / Impact:** TaxMate directly addresses the UMHackathon 2026 Domain 1 requirement — building a GLM-powered agentic workflow system that transforms fragmented, manual processes into intelligent automation. The solution aligns with Malaysia's national digitalisation agenda under MyDIGITAL, specifically the push for SME digital transformation and expanding tax compliance requirements. By integrating Z.AI's GLM as the central reasoning engine for document understanding, multi-step decision-making, and anomaly detection, TaxMate demonstrates how agentic AI can replace human coordination across an entire compliance pipeline. The platform is designed to scale across Malaysia's SME ecosystem, with a dual-sided marketplace model that simultaneously addresses the inefficiencies faced by both business owners and accounting professionals — and a roadmap that extends naturally into E-Invoice generation and submission as regulatory requirements evolve.

3. Product Purpose

3.1. Main Goal of the System

To provide an AI-powered tax compliance workflow platform that automates the end-to-end process of document processing and SST reporting for Malaysian SMEs — using Z.AI GLM as the central reasoning engine to eliminate manual coordination between business owners and accountants, while laying the architectural foundation for broader compliance coverage including E-Invoicing.

3.2. Intended Users (Target Audience):

- **SME Business Owners** — RM1M–5M revenue bracket, limited accounting knowledge, managing receipts manually. Primary need: upload receipts, track submission status, download signed SST forms, get AI tax guidance.
- **Accountants & Tax Agents** — Licensed professionals serving 8–12 SME clients, overloaded with data entry. Primary need: efficient multi-client review queue, AI-assisted verification, one-click sign-off.
- **Finance Managers** — In-house finance personnel responsible for monthly SST reporting. Primary need: centralised document management, AI-generated compliance insights.
- **Startup Founders** — Early-stage businesses navigating Malaysian tax obligations for the first time. Primary need: simple onboarding, AI chatbot for tax Q&A, guided compliance workflow.

4. System Functionalities

4.1. Description

TaxMate operates an intelligent, multi-agent tax compliance workflow engine powered by Z.AI GLM. When a business owner uploads a receipt or invoice, the system automatically orchestrates a seamless AI pipeline—from multi-modal OCR extraction to automated field classification and anomaly detection—eliminating repetitive manual data entry.

Accountants are presented with a prioritised review workspace where GLM's reasoning is fully transparent, allowing them to verify AI-processed data in minutes. The workflow concludes with a secure digital sign-off, where accountants provide a handwritten signature via a digital pad to authenticate and generate the final compliant SST documents. The entire process is stateful, ensuring GLM maintains context across every stage to adapt its reasoning based on document types, regulatory nuances, or detected anomalies.

4.2. Key Functionalities:

- **AI-Driven Document Perception & OCR:** Features an intelligent ingestion engine that accepts multi-format uploads (PDF, JPG, PNG) via batch drag-and-drop; utilizes AI-powered PaddleOCR for high-precision text recognition, which is then fed into the GLM-based Reasoning Agent to contextually interpret and structure unstructured data.
- **AI Field Extraction:** GLM classifies document types and extracts structured fields according to SST ledger formats with per-field confidence scoring. The extraction schema is designed to be extensible towards MyInvois E-Invoice fields as the platform evolves into a fully digital gateway.
- **Anomaly Detection:** A dual-layer checking system combining rule-based validation and GLM reasoning to flag issues such as invalid TIN formats, missing buyer information, or inconsistent amounts—providing plain-language explanations and suggested resolutions for each.
- **Secure Digital Sign-off Pad:** Replaces traditional manual processes with a handwritten digital signature interface. Once a review is complete, accountants provide a secure digital signature to authenticate the record, transitioning the document from a "Draft" status to a legally prepared tax record.
- **Intelligent Auto-Prioritisation:** Documents meeting or exceeding a 90% AI confidence threshold are automatically flagged as "Ready for Sign-off" to streamline batch processing. Documents with detected anomalies or low confidence are moved to the top of the accountant's queue for active intervention.
- **Accountant Review Workspace:** A professional three-panel interface displaying the original document image, editable AI-extracted results, and the GLM's full reasoning trace, enabling informed expert sign-off in minutes.
- **Compliant Document Generation:** Automatically produces SST-02 draft PDFs and generates a final signed SST form upon accountant approval, ready for LHDN submission.
- **Agent Thinking Panel:** A real-time streaming display of each AI agent's reasoning steps, tool calls, and logic outputs, making the GLM's decision process fully transparent and understandable for both owners and accountants.

- **AI-Assisted Tax Planning Suggestions:** GLM analyzes patterns across submitted documents to surface actionable insights—such as expense categories approaching deductible thresholds or recurring anomalies—presented as plain-language recommendations.
- **Owner AI Chatbot:** An embedded AI assistant in the Owner Dashboard providing real-time tax Q&A, SST classification guidance, and filing deadline reminders.
- **Accountant AI Chat Assistant:** A dedicated AI interface within the Accountant Workbench for querying tax rules and resolving ambiguous classifications without leaving the workspace.
- **AI Workflow Orchestrator:** A GLM-powered stateful orchestrator that monitors the entire 7-stage compliance pipeline — from document intake through to completion. The orchestrator applies adaptive gating logic to manage workflow progression: if more than 30% of documents in a submission carry risk flags, filing is automatically blocked until issues are resolved; if a filing deadline is approaching, the orchestrator fast-tracks the review queue; if total tax payable exceeds RM 10,000, additional verification is triggered before sign-off is permitted. In the event of GLM API unavailability, the orchestrator falls back to a deterministic rule-based engine — ensuring the system degrades gracefully without crashing or losing document state.

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

TaxMate leverages Ilmu-GLM-5.1—a specialized model developed through the collaboration between YTL AI Labs and Z.AI—as its core reasoning engine, selected for its unique optimization within the Malaysian financial landscape. Unlike general-purpose LLMs, Ilmu-GLM-5.1 excels at parsing the complex, mixed-language documents (English, Bahasa Malaysia, and Simplified Chinese) typical of local SMEs, ensuring high-precision data extraction from unstructured receipts. By utilizing its advanced function-calling capabilities, TaxMate orchestras a "stateful" multi-agent workflow that dynamically

triggers RAG-based SST rule-checkers and anomaly detection agents within a single reasoning chain.

4.3.2. Prompting Strategy:

TaxMate employs a multi-step agentic prompting strategy rather than zero-shot or few-shot approaches. Each document passes through a sequential chain of specialised agents — OCR Agent, Extraction Agent, Anomaly Agent, and Generation Agent — where each agent receives a focused prompt scoped to its specific task, along with the output of the previous step as context. This approach was chosen because tax document processing is inherently multi-stage: a single monolithic prompt attempting to extract fields, detect anomalies, and generate output simultaneously produces inconsistent results. Breaking the reasoning into discrete, auditable steps improves accuracy, makes errors traceable, and allows individual agents to be retried or replaced independently without affecting the entire pipeline.

4.3.3. Context & Input Handling:

TaxMate accepts document uploads up to 10MB per file with a maximum of 10 files per batch. For image-based inputs, the system utilizes AI-driven PaddleOCR to intelligently parse visual layouts and convert documents into high-fidelity raw text before GLM processing. This AI-enhanced extraction ensures that spatial relationships between fields are preserved, keeping token consumption predictable while maintaining data integrity. For text inputs exceeding the context window, the system applies semantic chunking—splitting documents by logical sections (header, line items, footer) rather than arbitrary character limits—to ensure extracted fields remain contextually coherent. Unparseable documents are flagged with specific AI-generated error reasons to prevent silent failures or partial outputs.

4.3.4. Fallback & Failure Behavior:

When GLM returns an unusable or low-confidence response, TaxMate follows a three-tier fallback mechanism. First, the system retries the prompt up to two times with a

reformulated instruction. If still unusable after retries, affected fields are marked with a low-confidence flag and surfaced prominently in the accountant's workspace for manual correction — ensuring the workflow continues rather than halting entirely. For critical failures such as complete extraction failure or hallucinated TIN numbers that fail format validation, the document is escalated to human review with a detailed error report. No output is ever silently accepted without passing at least basic rule-based validation.

5. User Stories & Use Cases

5.1 User Stories

Owner Stories:

- *"As an SME business owner, I want to select a tax period and upload all my receipts in one place so that my documents are organised by filing cycle without manual sorting."*
- *"As an SME business owner, I want to batch upload multiple receipts at once so that I do not have to upload documents one by one during a busy filing period."*
- *"As an SME business owner, I want to review all uploaded receipts before submitting so that I can ensure completeness and accuracy before handing over to my accountant."*
- *"As an SME business owner, I want the document set to lock upon submission so that the accountant reviews exactly what I submitted without post-submission changes."*
- *"As an SME business owner, I want to receive the signed SST form via email automatically so that I do not need to follow up with my accountant manually."*
- *"As an SME business owner, I want an AI chatbot on my dashboard so that I can ask tax-related questions and get instant guidance at any point in the process."*

- *"As an SME business owner, I want AI-generated tax planning insights on my dashboard so that I can make informed financial decisions throughout the year."*

Accountant Stories:

- *"As an accountant, I want to be notified by email when a client submits documents so that I can prioritise and manage my workload across multiple clients efficiently."*

- *"As an accountant, I want high-confidence documents to be auto-approved so that I can focus my attention on receipts that genuinely require manual verification."*

- *"As an accountant, I want to review each receipt individually with AI-extracted fields and GLM reasoning visible so that I can verify AI results and sign off with confidence."*

- *"As an accountant, I want AI assistance within my review workspace so that I can resolve ambiguous entries and query tax rules without leaving the platform."*

5.2 Use Cases (Main Interactions)

- **Owner Flow (Ingestion to Download):** Business owner selects tax period → uploads receipts/invoices via batch drag-and-drop → access Document Repository to verify uploaded vs. missing receipts → owner chooses to download Draft SST for internal review or initiates "Submit to Accountant" → system locks the submission to ensure data integrity → owner receives email notification upon accountant sign-off → visits Downloads section to access either the Draft version or the final Signed SST Form.

- **Accountant Flow (Review to Sign-off):** Accountant receives notification of client submission → click the client card to accesses Accountant Workbench to view the consolidated receipt list → three-panel workspace loads showing the original document image, GLM-extracted fields (editable), and the full Agent Thinking Panel displaying each reasoning step → once all receipts are reviewed, the platform surfaces a Digital Sign-board → accountant provides a handwritten signature → system instantly generates the signed

PDF form → automated email is triggered to deliver the finalized document back to the business owner.

5.3 Use Case Diagram

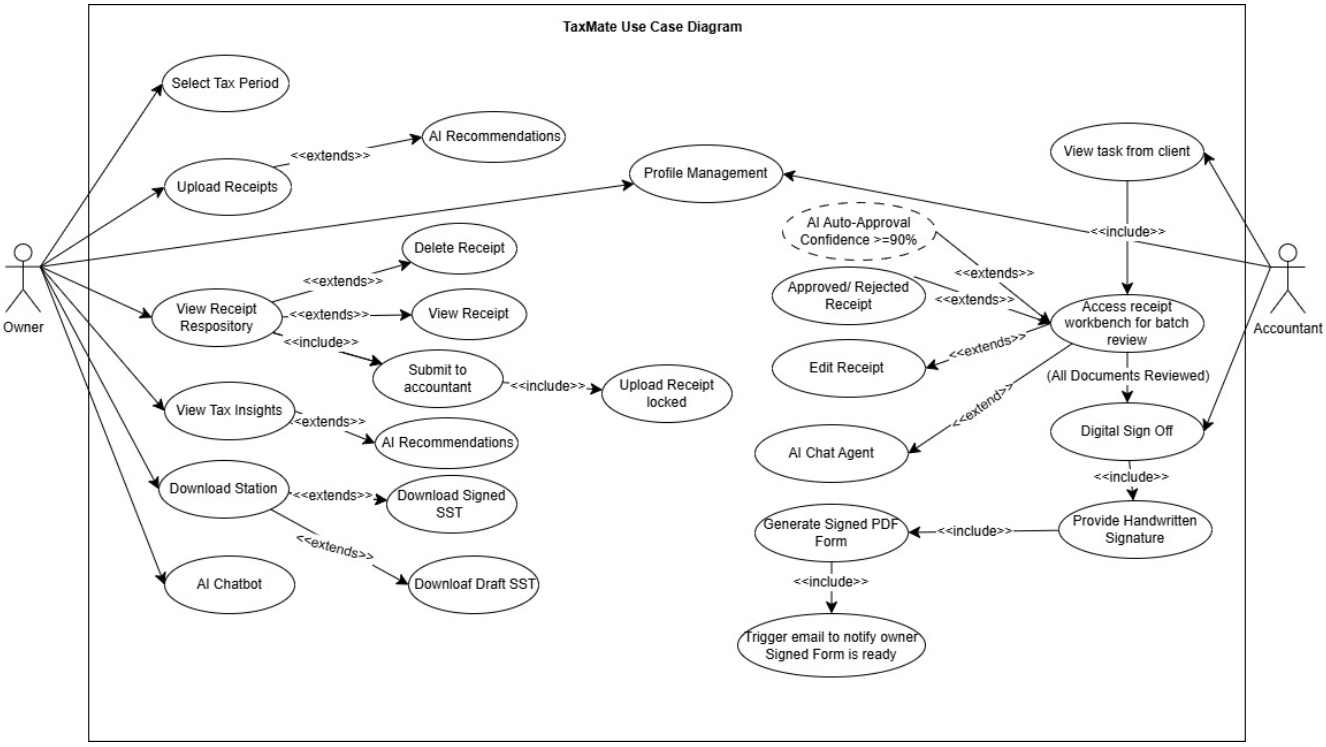

Figure 5.3.1 Use Case Diagram

6. Features Included (Scope Definition)

Features are classified into two tiers based on their criticality to the core compliance workflow and demo viability.

P0 — Core (must be present for the product to function):

- Dual-role platform supporting both SME business owner portal and accountant workspace. Rationale: without both roles, the collaboration model does not exist.
- Tax period selection with period-specific Document Repository. Rationale: core organisational structure of the compliance workflow.
- Batch document upload (PDF, JPG, PNG) with drag-and-drop functionality. Rationale: primary data ingestion mechanism for SME owners.
- Document lock upon owner submission. Rationale: ensures submission integrity before accountant review begins.
- AI-powered OCR and GLM field extraction with per-field confidence scoring. Rationale: central AI value proposition — without this, TaxMate is a file store.
- Intelligent auto-approval at 90% confidence threshold. Rationale: core efficiency driver that routes accountant attention to high-risk items only.
- Anomaly detection with plain-language explanations and resolution suggestions. Rationale: prevents compliance errors before they reach the accountant.
- Three-panel accountant review workspace with editable AI results and Agent Thinking Panel. Rationale: enables AI-assisted, explainable human review.
- Digital Sign-off via Handwritten Signature. Rationale: Completes the workflow with a legally binding approval and authenticates the finalized SST form.
- SST-02 draft PDF generation. Rationale: primary compliance output for the MVP.
- Signed SST form generation upon accountant sign-off. Rationale: final deliverable that validates the entire workflow.
- Automated email notifications for both accountants and business owners at key workflow stages. Rationale: closes the coordination loop without manual follow-up.
- AI Workflow Orchestrator with adaptive gating logic. Rationale: Stateful pipeline management ensures filing is blocked when risk thresholds are exceeded, deadlines are

respected, and high-value submissions receive additional scrutiny — preventing non-compliant documents from reaching sign-off.

P1 — Enhanced (differentiating features that strengthen the product):

- Owner AI Chatbot embedded in the Owner Dashboard. Rationale: real-time tax guidance reduces owner errors and increases platform stickiness.
- Accountant AI Chat Assistant embedded in the Accountant Workbench. Rationale: accelerates review by surfacing tax rules contextually within the workspace.
- AI-assisted tax planning suggestions surfaced within the business owner dashboard. Rationale: adds proactive value beyond reactive compliance processing.
- Statistics dashboard with AI-generated insights for both owner and accountant roles. Rationale: provides longitudinal compliance visibility and actionable intelligence.

7. Features Not Included (Scope Control)

The following features are explicitly out of scope for the current MVP. Each exclusion is a deliberate scope decision to maintain development focus while preserving a clear product roadmap.

- **Direct API submission to LHDN MyInvois portal** — Requires LHDN developer approval; listed as primary roadmap milestone post-MVP.
- **Full MyInvois E-Invoice generation (55-field JSON output)** — Architectural groundwork included in extraction schema; full implementation is a roadmap milestone.
- **Multi-company or multi-branch account support** — Single-entity scope for MVP; multi-entity support planned for enterprise tier.
- **EPF, SOCSO, or payroll module integration** — Out of current compliance scope; roadmap item for HR-adjacent expansion.
- **Real-time WhatsApp notification delivery** — Message preview shown in demo; actual delivery requires WhatsApp Business API approval.
- **Accountant credential verification** — Professional certification validation is a trust-and-safety feature planned for marketplace launch.

8. Assumptions & Constraints

LLM Cost Constraint: Each document processed through TaxMate's full agent pipeline consumes an estimated 2,000–4,000 tokens — covering OCR text input, field extraction, anomaly detection, and output generation. TaxMate implements RAG-based caching for frequently referenced tax rules (SST regulations, MSIC codes), so repeated lookups do not re-query GLM. Agent prompts are kept concise and task-scoped to avoid unnecessary context bloat across the pipeline.

Technical Constraints: Document processing accuracy is dependent on input image quality; low-resolution or heavily compressed images may result in incomplete OCR output. The system currently supports Bahasa Malaysia, English, and simplified Chinese text within documents.

Performance Constraints: Full agent pipeline processing per document takes an estimated 15–30 seconds depending on document complexity and file size. Batch uploads are processed asynchronously via Celery task queue, so users are not blocked waiting for results.

Human-in-the-Loop Requirement: GLM extraction results require accountant review and sign-off before any generated document is considered final. The system is not designed for fully unsupervised production submission without human approval in the loop.

Auto-Approval Threshold Assumption: The 90% confidence threshold for auto-approval is calibrated based on internal testing with sample Malaysian SME receipts. This threshold may be adjusted based on real-world accuracy data collected post-launch.

Regulatory Assumption: TaxMate assumes the current SST regulatory framework remains the primary compliance obligation for the RM1M–5M SME segment during the MVP phase. E-Invoice mandate expansion timelines are monitored as a trigger for roadmap acceleration.

9. Risks & Questions Throughout Development

The following risk register covers technical, operational, compliance, and product risks identified throughout TaxMate's development.

Risk 1 — OCR Accuracy on Poor-Quality Receipts Description: Handwritten, faded, or non-standard receipts may yield incomplete OCR output, leading to extraction failures or incorrect field

values. Mitigation: Low-confidence extractions are flagged for mandatory human review. Users receive specific error feedback rather than silent failures. Document quality guidelines are provided at the upload stage. Severity: Medium

Risk 2 — GLM Hallucination on Critical Tax Fields Description: GLM may generate plausible but factually incorrect TIN or SST registration numbers that appear valid to the accountant. Mitigation: All critical fields pass through rule-based format validators post-extraction. Hallucinated values failing validation are flagged before reaching the accountant. Dual-layer validation (rule-based + GLM) reduces single-point failure risk. Severity: High

Risk 3 — Auto-Approval of Confidently Wrong Extractions Description: Documents above the 90% threshold may still contain factually incorrect fields that bypass manual review entirely. Mitigation: Auto-approved items remain visible in the accountant queue for spot-checks. The audit trail records confidence score and approval method for every field. Accountants can override auto-approvals at any time. Severity: High

Risk 4 — Accountant Review Queue Bottleneck Description: High simultaneous document volumes may create an unmanageable review backlog for accountants serving many clients. Mitigation: Auto-approval at 90% confidence significantly reduces manual review volume. Priority sorting surfaces anomalous and high-value documents first. The accountant dashboard provides workload visibility across all clients. Severity: Medium

Risk 5 — Document Lock Misuse Description: Business owners may lock submissions prematurely without verifying all documents, creating re-submission friction and delaying the accountant workflow. Mitigation: A pre-submission checklist and document count with thumbnail preview are shown to the owner before the lock is confirmed. Accountants can flag incomplete submissions for owner re-submission. Severity: Low

Risk 6 — Email Notification Delivery Failure Description: Critical workflow notifications — new submission alerts and signed form delivery — may fail due to spam filtering or email server

issues. Mitigation: In-platform notifications are shown alongside email at every trigger point. Notification status is visible in both dashboards. A retry mechanism handles failed email delivery. Severity: Low

Risk 7 — Regulatory Change Risk Description: Changes to SST regulations, MSIC codes, or E-Invoice requirements may invalidate the RAG knowledge base or extraction schema. Mitigation: The RAG knowledge base is modular and updatable independently of the agent pipeline. A regulatory monitoring process is defined for post-MVP maintenance cycles. Severity: Medium